

# Static Gesture Classification — Run Report

Generated 2026-05-13 11:17:42

## Data paths

|                       |                                                                         |
|-----------------------|-------------------------------------------------------------------------|
| Training data path 1  | /home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic · 35 trials       |
| Training data path 2  | /home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic · 35 trials       |
| Training data path 3  | /home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic · 36 trials     |
| Training data path 4  | /home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic · 35 trials     |
| Training data path 5  | /home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic · 35 trials      |
| Training data path 6  | /home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic · 24 trials     |
| Training data path 7  | /home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic · 35 trials     |
| Training data path 8  | /home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic · 35 trials    |
| Training data path 9  | /home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic · 36 trials |
| Training data path 10 | /home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic · 35 trials    |
| Total training paths  | 10 · 341 trials total                                                   |
| Report output dir     | /home/jestin/ThesisRepo/ML/NewReports/DynamicReports                    |

## Sensor selection

|                |                                                                                                             |
|----------------|-------------------------------------------------------------------------------------------------------------|
| Hands          | left                                                                                                        |
| Segments       | palm_mid, thumb, index, middle, ring, pinky, wrist                                                          |
| Modalities     | Accel                                                                                                       |
| Sensor columns | 39 channels                                                                                                 |
| Classes        | 7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll |

## Preprocessing, features & split

|                   |                                                                 |
|-------------------|-----------------------------------------------------------------|
| Resample steps    | 90                                                              |
| Butterworth       | off · cutoff 10.0 Hz · order 4 · fs 30.0 Hz                     |
| Normalisation     | minmax                                                          |
| Feature mode      | stats+fft → feature dim 2106                                    |
| Split / seed / CV | random_state=42 · CV: LOUO (10 folds, every user held out once) |
| Sample counts     | train (post-aug)=1705 · LOUO-evaluated samples=1705             |

## Augmentation (training only)

|                        |                        |
|------------------------|------------------------|
| Enabled                | yes                    |
| Copies per train trial | 4                      |
| Jitter                 | off · sigma=0.01       |
| Amplitude scale        | on · range=(0.9, 1.1)  |
| Time warp              | off · range=(0.9, 1.1) |

## Classifiers — LOUO cross-validation

| Algorithm     | CV mean (LOUO, 10 folds) | CV std | CV min | CV max |
|---------------|--------------------------|--------|--------|--------|
| Random Forest | 0.8787                   | 0.0737 | 0.7714 | 1.0000 |

## Per-class CV F1 (across all LOUO folds)

| Class                | Random Forest | Support |
|----------------------|---------------|---------|
| Double_Lower         | 0.923         | 250     |
| Double_Nothing       | 0.676         | 230     |
| Double_Pistol_Recoil | 0.845         | 250     |
| Double_Raise         | 0.913         | 250     |
| Double_Wiggle        | 0.957         | 245     |
| Double_cmere         | 0.938         | 255     |
| Drum_Roll            | 0.840         | 225     |
| macro avg            | 0.870         | 1705    |

## Per-user accuracy (each user as held-out fold)

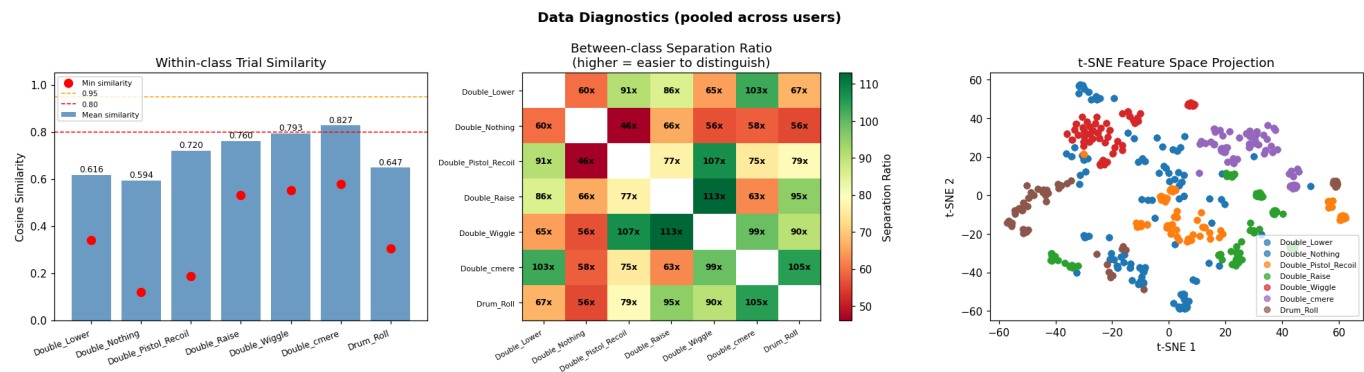
| User         | n trials | Random Forest |
|--------------|----------|---------------|
| 1_Alan       | 175      | 1.000         |
| 2_Alex       | 175      | 0.800         |
| 3_Anghad     | 180      | 0.928         |
| 4_Daniel     | 175      | 0.886         |
| 6_Harry      | 175      | 0.937         |
| 8_Jestin     | 120      | 0.958         |
| 11_Mansh     | 175      | 0.794         |
| 12_Marcus    | 175      | 0.771         |
| 14_StephenV2 | 180      | 0.883         |
| 17_Raquel    | 175      | 0.829         |

|                     |  |       |
|---------------------|--|-------|
| Mean (across users) |  | 0.879 |
| Std (across users)  |  | 0.074 |

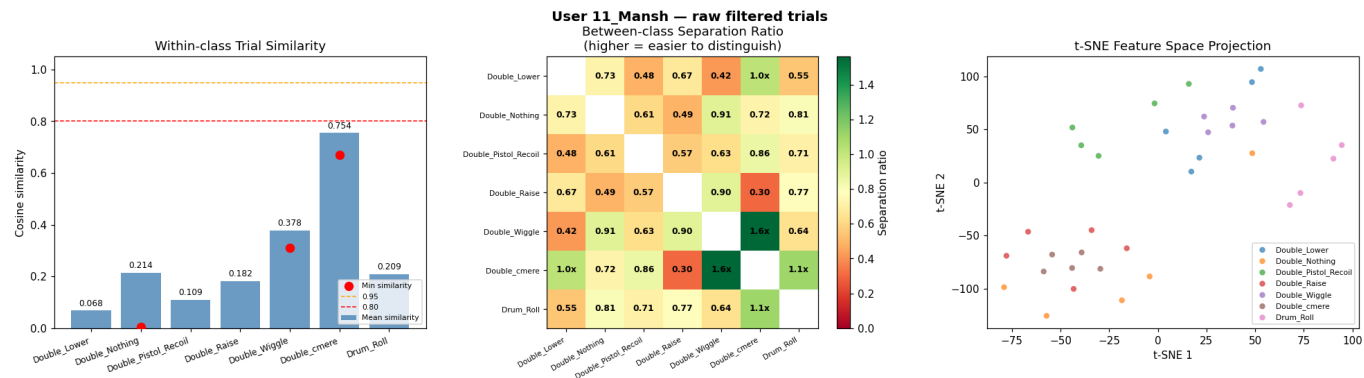
Appendix — Data Diagnostics

21 figures: pooled across users + one per detected user (raw filtered trials and extracted-feature views).

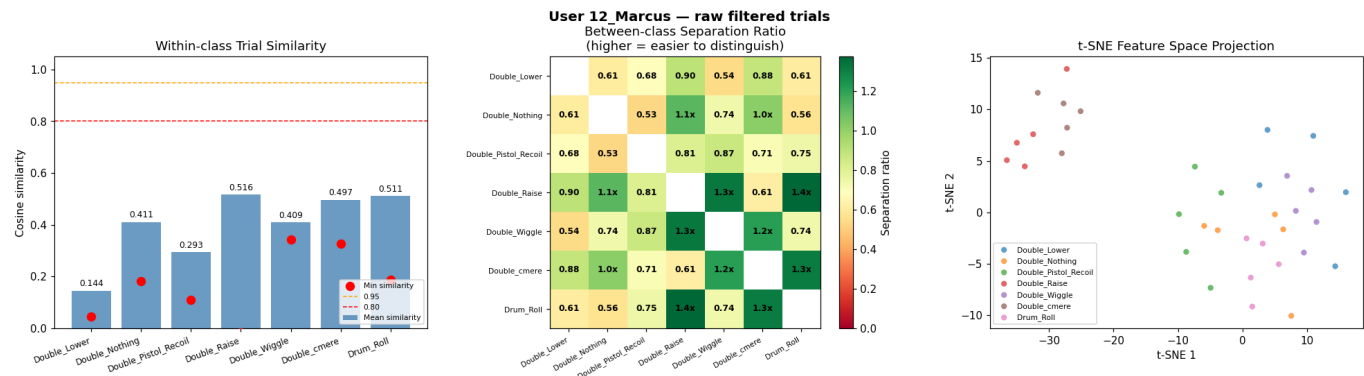
Pooled diagnostics — extracted features (across all users)



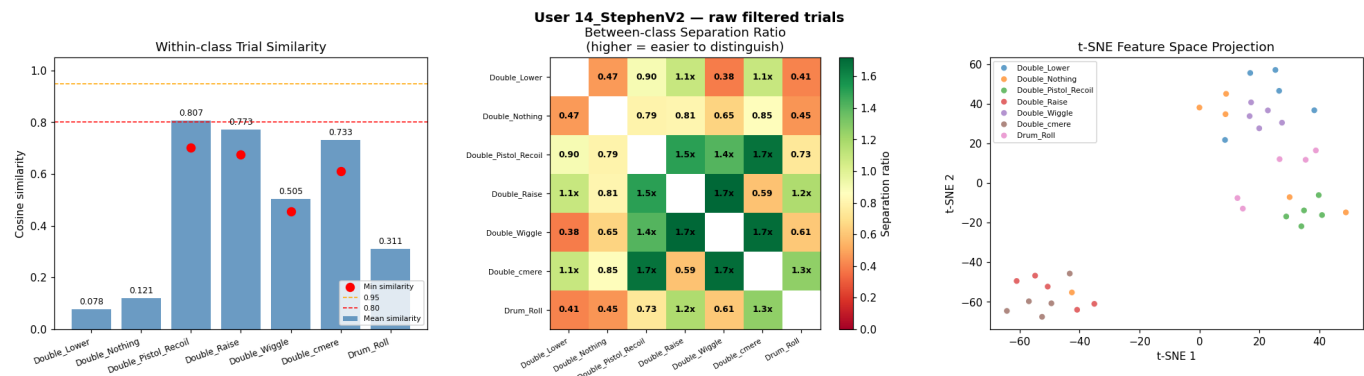
User 11\_Mansh — raw filtered trials



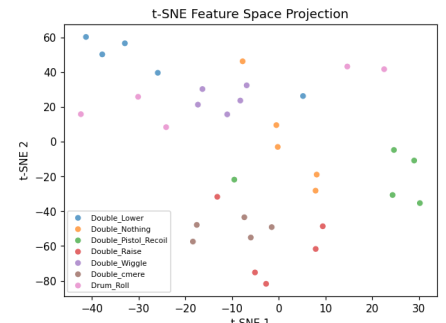
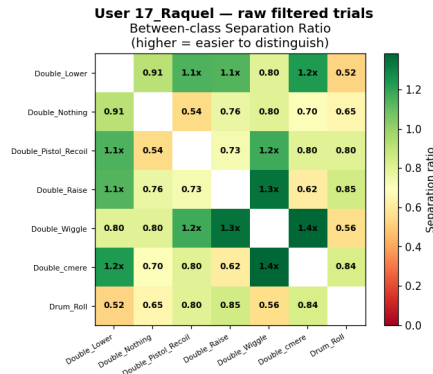
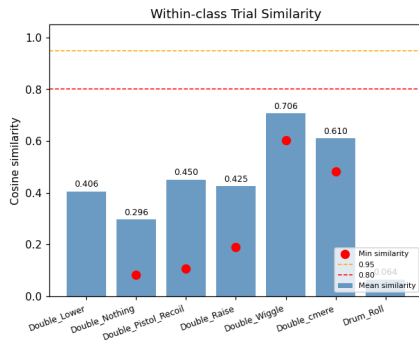
User 12\_Marcus — raw filtered trials



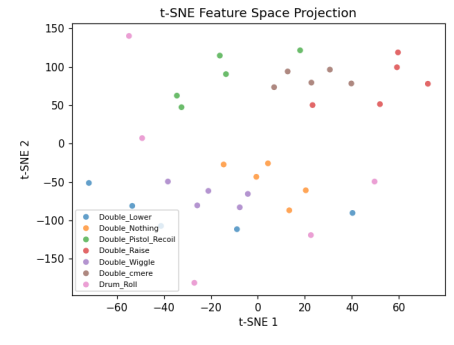
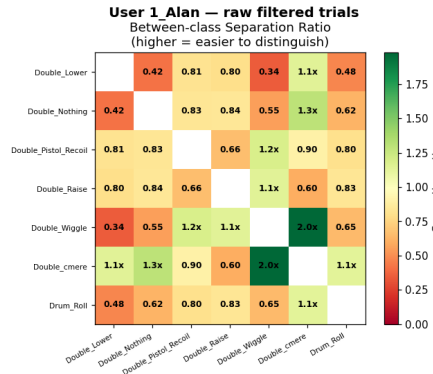
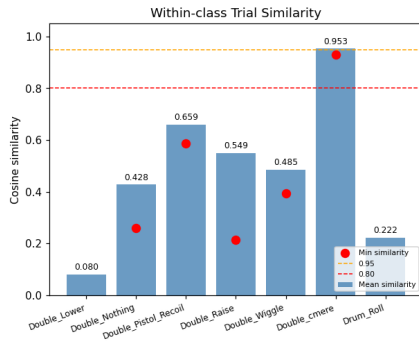
User 14\_StephenV2 — raw filtered trials



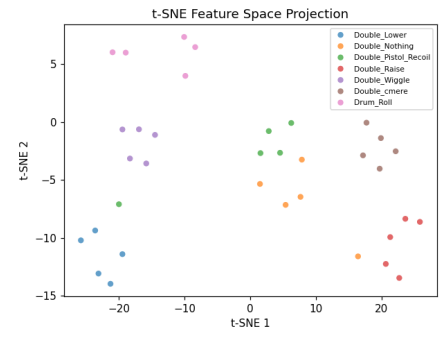
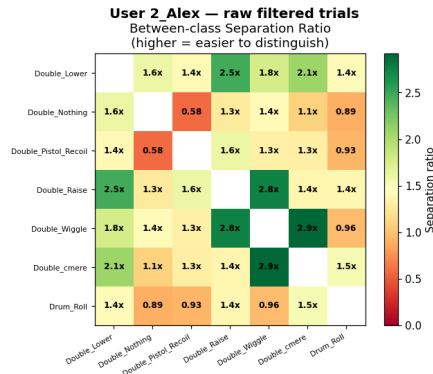
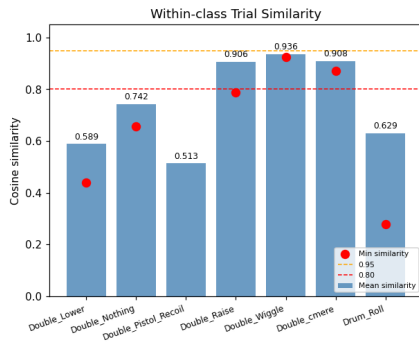
User 17\_Raquel — raw filtered trials



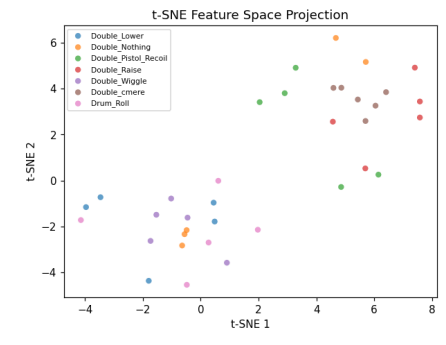
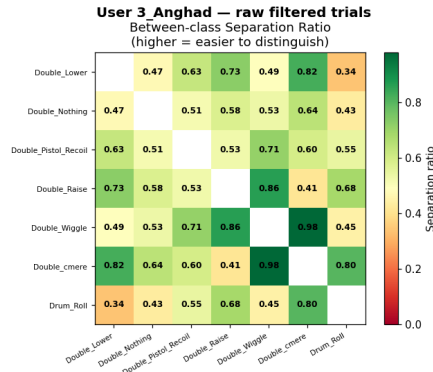
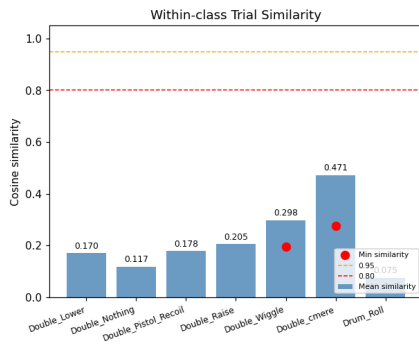
## User 1\_Alan — raw filtered trials



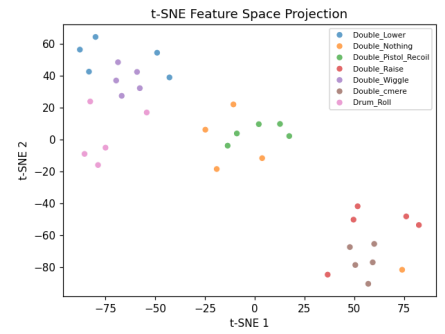
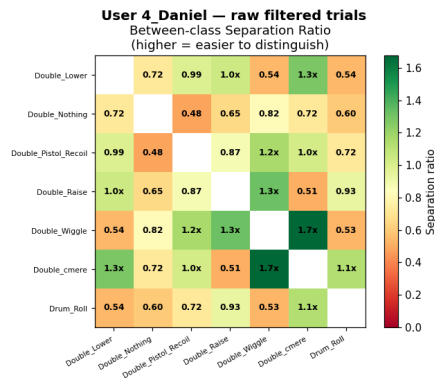
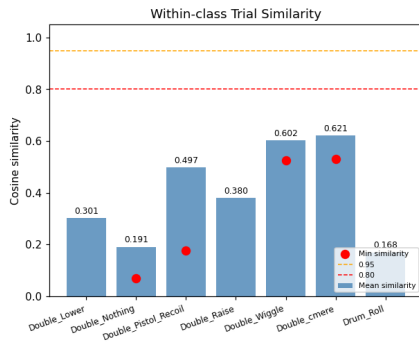
## User 2\_Alex — raw filtered trials



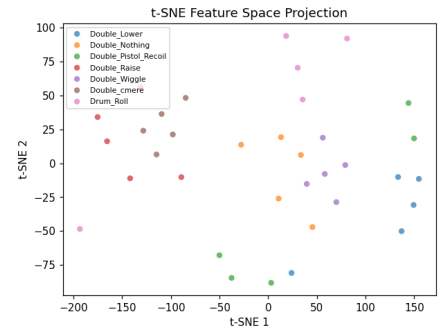
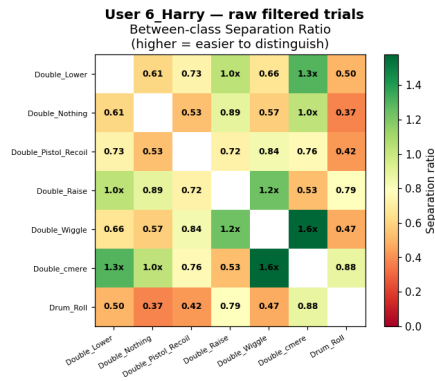
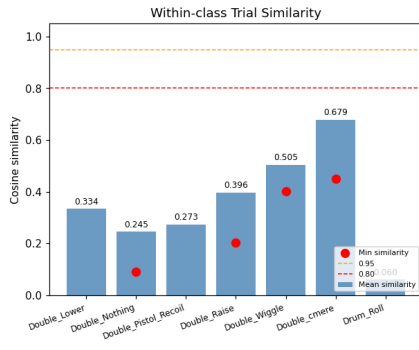
## User 3\_Anghad — raw filtered trials



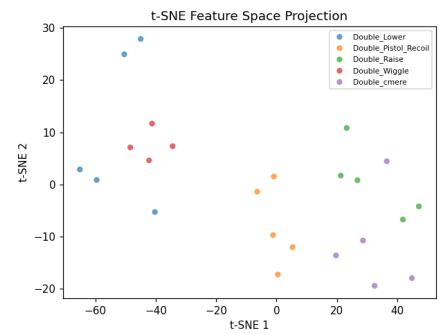
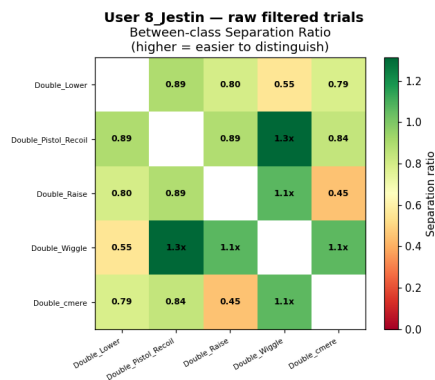
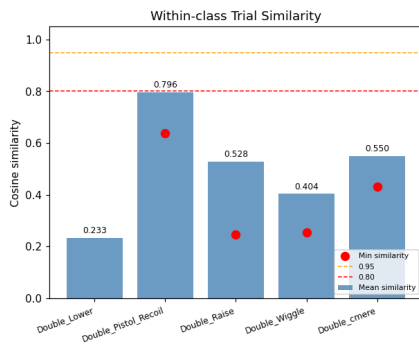
## User 4\_Daniel — raw filtered trials



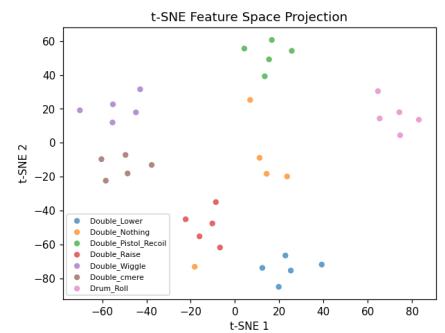
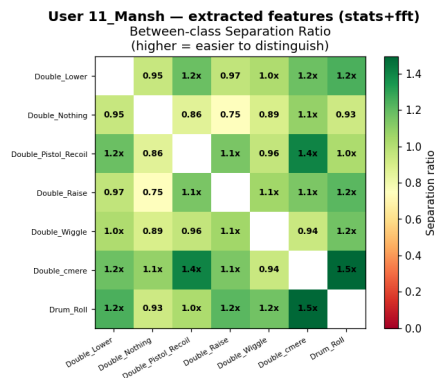
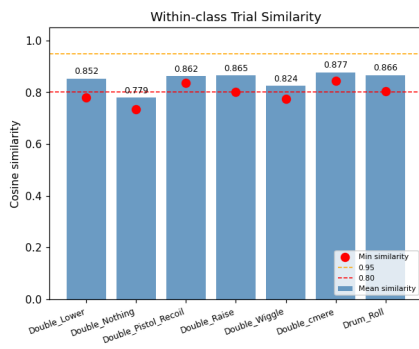
## User 6\_Harry — raw filtered trials



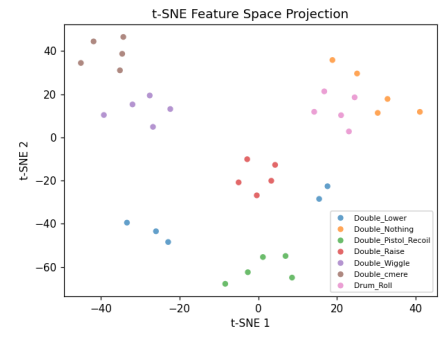
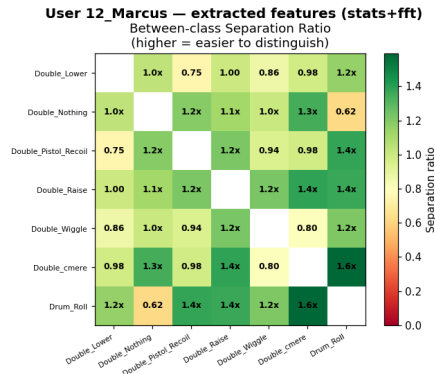
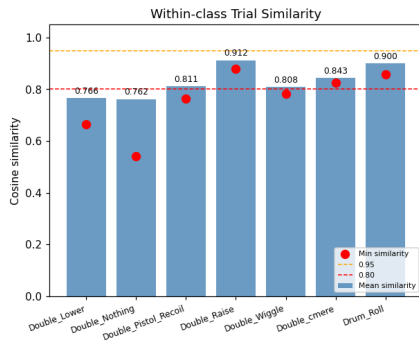
## User 8\_Jestin — raw filtered trials



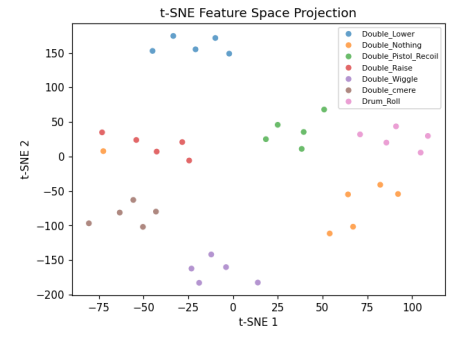
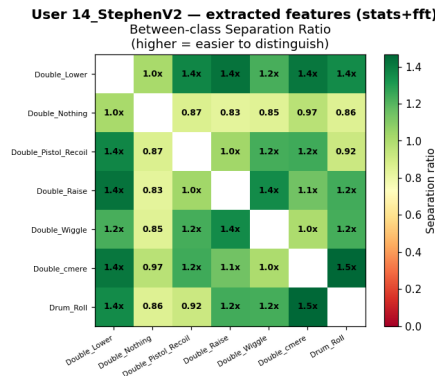
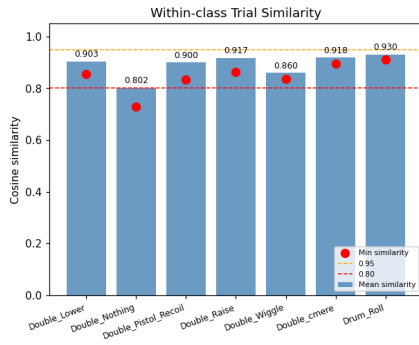
## User 11\_Mansh — extracted features (stats+fft)



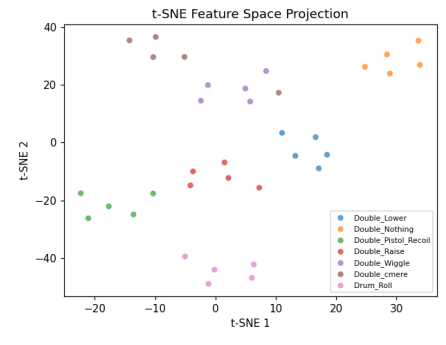
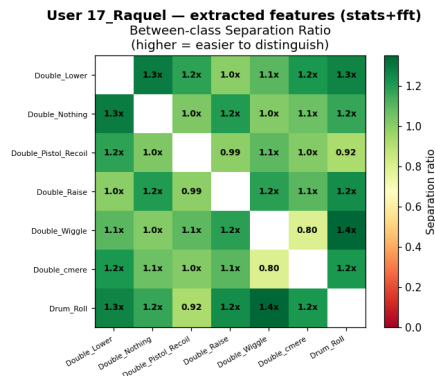
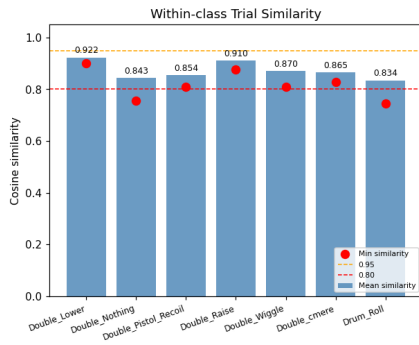
## User 12\_Marcus — extracted features (stats+fft)



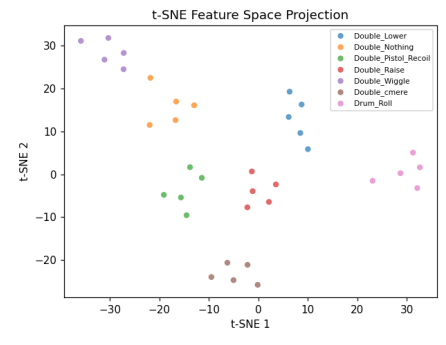
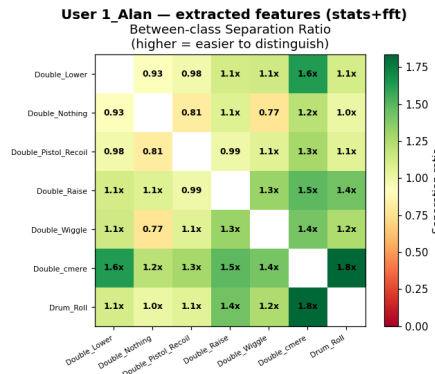
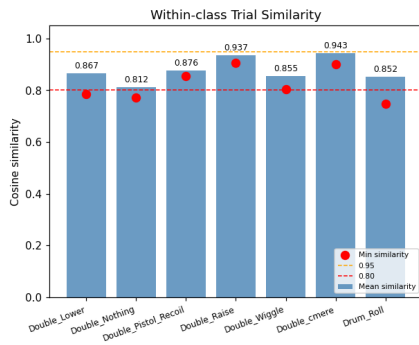
## User 14\_StephenV2 — extracted features (stats+fft)



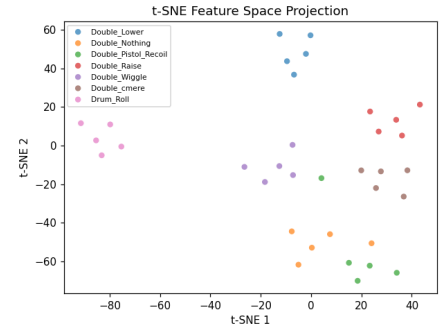
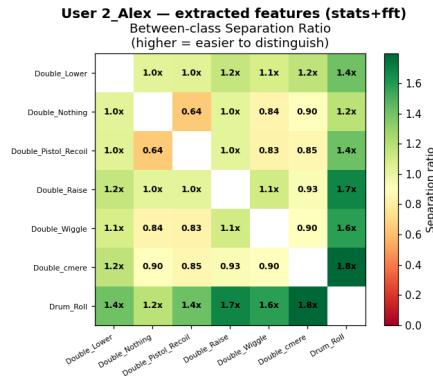
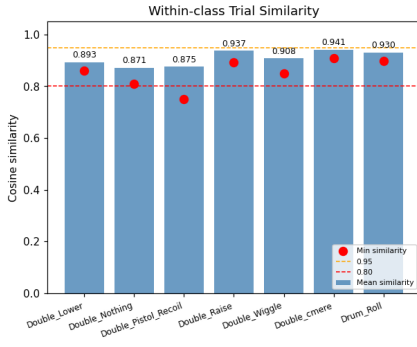
## User 17\_Raquel — extracted features (stats+fft)



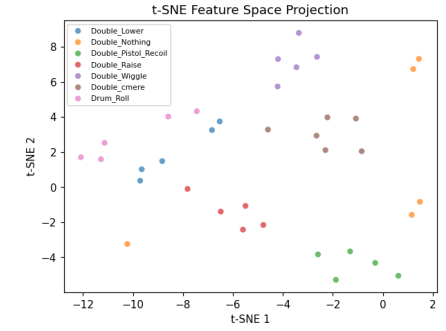
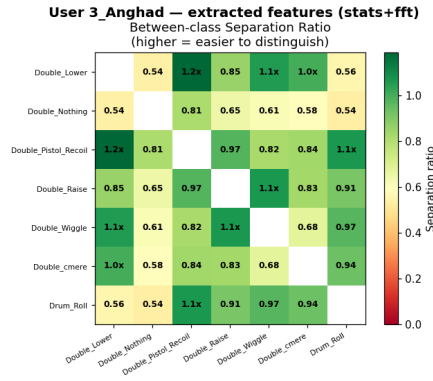
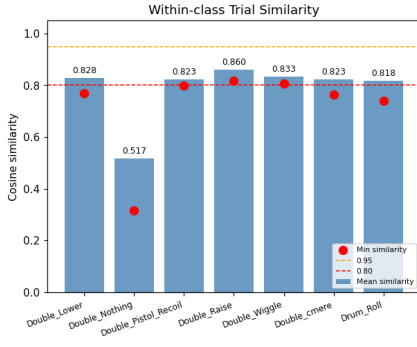
## User 1\_Alan — extracted features (stats+fft)



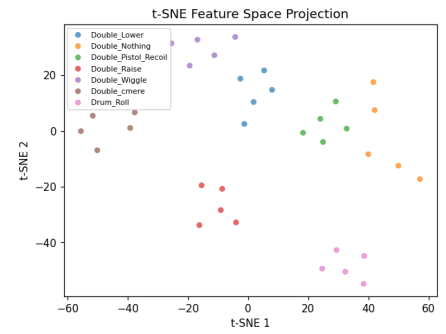
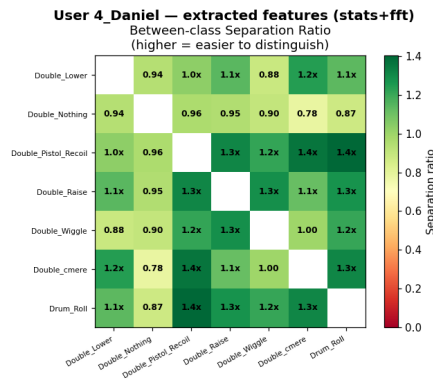
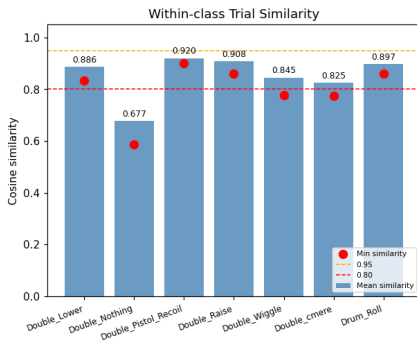
## User 2\_Alex — extracted features (stats+fft)



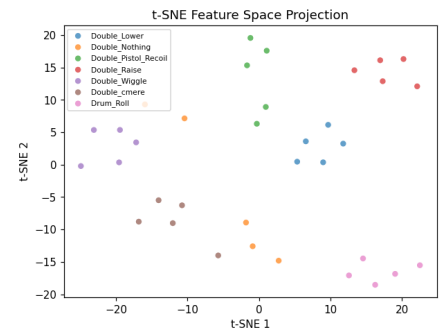
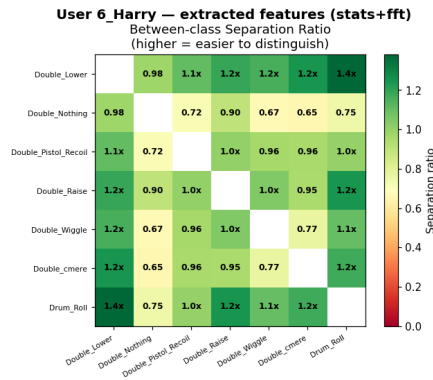
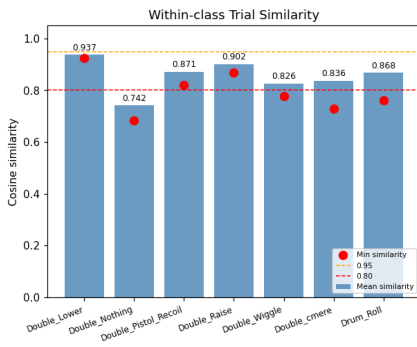
### User 3\_Angad — extracted features (stats+fft)



### User 4\_Daniel — extracted features (stats+fft)



### User 6\_Harry — extracted features (stats+fft)



### User 8\_Jestin — extracted features (stats+fft)

